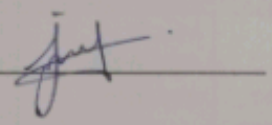
SIMATS ENGINEERING
ASSESSMENT TOOL 3: Application Based Questions

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1519

COURSE OUTCOME COVERED
CO2: Analyze virtualization architectures to identify performance and security issues in cloud computing environments. (BL3)

Dave's Taxonomy: Manipulation (Level 2) Total Marks: 25
Question Count: 5

Code of Conduct
I Jannathul Firdaus.N (192573025) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: 

Assessment Tasks

Application Based Questions

1. A company experiences unpredictable traffic spikes during seasonal sales. Explain how cloud auto-scaling and load balancing can handle this demand efficiently. How does this approach reduce downtime and improve user experience? What role do monitoring tools play in this setup? Relate your answer to a real-world e-commerce scenario.

2. An organization wants to store large volumes of backup data securely in the cloud.
   Discuss how cloud storage solutions ensure durability and data redundancy.
   How can encryption and access control protect sensitive backup data?
   Explain the cost advantages compared to traditional storage systems.
   Provide an example of a disaster recovery use case.

3. A mobile app company plans to deploy its application globally. Explain how content delivery networks (CDNs) improve performance and reduce latency. How do edge servers enhance user experience across regions? Discuss how cloud providers support global availability. Relate this to a real-time streaming or gaming application.

4. A financial institution requires strict compliance and data security in cloud usage. Explain how cloud providers support regulatory compliance and auditing. What role do logging, monitoring, and encryption play in security? How can the company ensure data sovereignty requirements are met? Give an example involving sensitive financial transactions.
5. A development team wants to adopt containerization for application deployment. Explain how containers differ from virtual machines in terms of performance and resource usage. How does orchestration (like Kubernetes) simplify deployment and scaling? Discuss portability benefits across different cloud environments. Provide an example of a microservices-based application.

Application Based Questions Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Relevance to Application	25%	Response directly addresses the given use case with strong relevance	Mostly relevant response	Partially relevant	Weak relevance
Use of Cloud Concepts	25%	Applies appropriate concepts accurately	Mostly accurate application	Basic application only	Incorrect concept usage
Practical Insight	20%	Shows strong real-world understanding	Shows reasonable practical understanding	Limited practical insight	No practical insight
Completeness	15%	Response covers all required aspects	Covers most aspects	Covers only basic aspects	Incomplete response
Clarity	15%	Clear and concise answer	Generally clear	Some confusion in expression	Poor clarity

1. Auto Scaling and Load Balancing During Seasonal Sales

Introduction:

Auto Scaling and load balancing help cloud applications handle sudden increases in traffic. They improve performance, maintain availability, reduce downtime and optimize resource usage.

Cloud Auto Scaling:

* Scale-out - Adds instances when traffic increases
* Scale-in - Remove unnecessary instances when traffic decreases.
* Predictive scaling - use historical patterns to prepare for expected traffic.
* Scheduled scaling - Adds resources at predefined times such as festival sales.

Load Balancing:

It distributes incoming requests across multiple healthy servers. It prevents server overload and redirects traffic away from failed instances, improving response time and availability.

Real-world E-commerce Scenario.

During Festive Sale, an e-commerce website experiences thousands of simultaneous visitors. Auto-Scaling adds servers, while a load balancer distributes customer requests among them. monitoring tools detect increasing demand and trigger scaling actions. After the Sale unnecessary instances are removed to reduce costs.

Impact on Availability and User Experience:

These techniques reduce overload-related failures improve page loading speed and allow customers to complete purchases smoothly Redundant infrastructure and resilient databases further improve availability.

2. Secure cloud storage for Backup data.

Introduction:

cloud Storage allows organizations to store large amounts of backup data without maintaining expensive physical infrastructure. Redundancy, encryption and access control help protect important information.

Durability and Data Redundancy:

* Replication - maintains multiple copies of data.
* geographic redundancy - Stores multiple copies across separate locations.
* Versioning - Preserves earlier version of files.
* integrity checks - Detects data corruption.
* Lifecycle policies - moves older backups to lower cost storage tiers

Encryption and Access control

Encription at rest: protects stored data from unauthorized reading.

Encryption in transit: Uses protocols such as TLS to secure data during transfer

Access control - IAM policies restrict access to authorized users. Additional measure include multi factor authentication least privilege permissions, encryption key management.

Cost advantages compared with traditional Storage:

→ Pay as you go: pay according to storage usage and selected services.

→ Lower hardware costs: Reduces the need to purchase physical storage device.

→ Reduced maintenance: The provider manages the underlying infrastructure.

→ flexible capacity: Storage can grow without installing additional hardware

→ Archival Storage: infrequently accessed backups can be stored at lower rates.

Disaster Recovery Scenario:

A hospital stores patient records and applications backups in cloud storage. if a fire destroys its primary data centre

the hospital can restore its systems using protected backups stored in a separate location.

The hospital retrieves the latest valid backup restores its applications and databases and checkups data integrity before resuming operations. A defined recovery point objective (RPO) limits acceptable data loss while a recovery Time objective (RTO) defines the target restoration time.

3. CDNs and Global Deployment of a Mobile Application:

Introduction:

A content Delivery Network (CDN) is a distributed network of sanor that delivers content from location close to users. It reduces latency, improves loading speed, and supports applications serving users across different contries.

How CDNs Reduce Latency:

CDNs cache frequently accessed content such as image, videos, and applications files on nearby edge servers.

When a user requests cached content, It is delivered from the nearest suitable CDN location instead of repeatedly contacting the origin server. This reduces response time network traffic and origin server workload.

Role of Edge Server:

- Faster content loading
- Reduced buffering and delays
- Lower traffic to origin servers.
- Better handling of sudden traffic increases
- Support for certain types of processing close to users.

How Global Availability is provided by cloud.

* Regional deployment - places applications closer to users.

- Global traffic routing - Directs requests to suitable endpoints.
- Health checks and failover: Redirects traffic away from unhealthy points
- Data replication: Maintains copies of data in supported locations.
- monitoring : Tracks latency, availability and errors.

Real Time Example: Gaming Application:

A global gaming company uses a CDN to distribute game updates, maps, and textures to players in different countries. Nearby edge servers deliver these files quickly reducing download time and network traffic.

The company also deploys regional game servers to reduce gameplay latency. Cloud auto scaling handles increases in player numbers while monitoring and health checks help detect failures.

4. Regulatory compliance and Data security in cloud computing:

Introduction:

financial institutions process sensitive information including account details and financial transactions. cloud security must protect this data while supporting applicable regulatory, auditing and data location requirements.

Regulatory compliance and Auditing:

cloud providers offer compliance certifications independent audit reports, and security services that help organizations meet regulatory requirements.

Financial institutions should review provider certifications, define security responsibilities maintain audit evidence and conduct regular risk assessments. provider certifications alone does not guarantee that every applications is compliant.

Role of Logging, Monitoring and Encryption:

Logging: Records login attempts, access to sensitive information, configuration changes and transactions-related events.

Monitoring: Analyzes security events and identifies suspicious activities or such as repeated failed logins or unusual transactions.

Encryption: protects data at rest and in transit.

Ensuring Data Sovereignty:

- Selecting approved cloud regions
- Verifying the locations of backups, replicas and logs.
- Reviewing cross-border data transfer rules.
- Applying access restrictions and contractual controls.
- Auditing compliance with applicable regulations.

Real world Example: Online Banking Transactions:

A bank processes fund transfer through a cloud-hosted application. It encrypts transaction data, authenticates users and restricts access through IAM policies.

Audit logs record transaction-related activities while monitoring tools identify suspicious transfer and unusual login attempts. The bank stores regulated data in approved locations and verifies that backups and replicas also meet location requirements.

5. Containers, VM, Kubernets orchestration:

Introduction:

containerization packages applications and their dependencies into portable units called containers. Containers and virtual machines support application deployment in different ways while kubernets automates the management and scaling of containerized applications.

containers Vs VM.

feature	containers	Virtual Machines
Architecture	share the host OS kernel	Run a separate guest OS
Resource Usage	generally light-weight	generally use more resources.
Startup time	Usally faster	usually slower
isolation	process level isolation.	separate guest OS isolation boundary
Protability	Runs on compatible container environments	Run on compatible virtualization platforms.
common Use	Microservices & APIs	Legacy and OS specific applications.

How Kubernets simplifies deployment and scaling.

* Automated deployment - deploys containarized application s across cluster nodes.
* Scaling - Adjust the no of applications according to demand
* Self healing - Restarts failed containers and replaces unhealthy pods.
* Rolling updates - Release new versions.

Real world Example:

An E-commerce platform contains separate services for every management. Each service run in its own container.

Kubernets manages these services and scales them independently. During a festival sale the product catalog service can increase its replicas to handle more requests while the payment service scales according to its own workload.

The same container images can be deployed across compatible cloud environment with appropriate infrastructure.